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SARDAR PATEL UNIVERSITY

External Examination – December 2023

BCA - Semester-I (New Course) (NEP 2020)

[US01MABCA01] : Programming Fundamentals Using C



Date: 4/12/2023 (Monday)

Time: 10.00am to 12.30pm

Total Marks: 50

Note: Figure indicates at right is maximum marks for question.

Q -1 Select an appropriate option for the following statements. [10]

1. _____ is a pictorial representation of an algorithm.
[a] Flow Chart [b] Data Diagram
[c] Pie Chart [d] Program
2. Translator which converts high-level language programs into machine language program is called _____.
[a] Compiler [b] Interpreter
[c] Assembler [d] Translator
3. C language is _____ language.
[a] High Level [b] Low Level
[c] Machine Level [d] Assembly
4. The value that can be changed during program execution is known as _____.
[a] Variable [b] Constant
[c] Operator [d] None of these
5. _____ section is compulsory in C program.
[a] main function [b] Definition section
[c] documentation [d] None of these
6. _____ format specifier used for integer data type.
[a] %d [b] %f
[c] %c [d] %e
7. _____ loop executes at least once.
[a] while [b] do...while
[c] if [d] for
8. Array subscript (index) begins with _____.
[a] Zero [b] One
[c] Two [d] None of these
9. Ampersand (&) symbol is not required to read _____ value in scanf().
[a] int [b] float
[c] string [d] All of above
10. '\0' is called _____.
[a] Null character [b] Character value
[c] Escape sequence [d] Symbolic constant

Q:2 What is an algorithm? Write characteristics of an algorithm. Also write an algorithm to Calculate Simple Interest (where, $SI = PRN/100$) and draw the flow chart for the same. [10]

OR

Q:2 What is Computer Language? List and explain generation of computer languages. [10]

Q:3 Draw and explain basic structure of C program with suitable program. [10]

OR

Q:3 What is an operator? List and explain any four operators with example. [10]

Q:4 Explain looping statements (Iterative control structures) with syntax and example. [10]

OR

Q:4 Define Array and 1-D Array. Explain declaration and initialization of 1-D array with syntax and example. [10]

Q:5 What is String? List all and explain any four string handling functions with syntax and example. [10]

OR

Q:5 List and explain functions of <stdio.h> header files with syntax and example [10]
